

Homework 3

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Problem 1

(a) Given

$$ds^2 = d\theta^2 + \sin^2 \theta d\phi^2,$$

we have

$$g_{\mu\nu} = \begin{pmatrix} 1 & 0 \\ 0 & \sin^2 \theta \end{pmatrix}.$$

The Lagrangian is given by

$$\mathcal{L} = g_{\mu\nu} \frac{dx^\mu}{d\sigma} \frac{dx^\nu}{d\sigma} = \dot{\theta}^2 + \sin^2 \theta \dot{\phi}^2.$$

Here we denote the σ -derivatives with Newton's notation. This satisfies the Euler-Lagrange equations

$$\frac{d}{d\sigma} \frac{\partial \mathcal{L}}{\partial \dot{x}^\mu} - \frac{\partial \mathcal{L}}{\partial x^\mu} = 0.$$

The partials are

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial \theta} &= 2 \sin \theta \cos \theta \dot{\phi}^2, & \frac{\partial \mathcal{L}}{\partial \phi} &= 0, \\ \frac{\partial \mathcal{L}}{\partial \dot{\theta}} &= 2\dot{\theta}, & \frac{\partial \mathcal{L}}{\partial \dot{\phi}} &= 2 \sin^2 \theta \dot{\phi}. \end{aligned}$$

We then have

$$\frac{d}{d\sigma} \frac{\partial \mathcal{L}}{\partial \dot{\theta}} = 2\ddot{\theta}, \quad \frac{d}{d\sigma} \frac{\partial \mathcal{L}}{\partial \dot{\phi}} = 4 \sin \theta \cos \theta \dot{\theta} \dot{\phi} + 2 \sin^2 \theta \ddot{\phi}.$$

This yields the system

$$\begin{aligned} 4 \sin \theta \cos \theta \dot{\theta} \dot{\phi} + 2 \sin^2 \theta \ddot{\phi} &= 0, \\ 2\ddot{\theta} - 2 \sin \theta \cos \theta \dot{\phi}^2 &= 0. \end{aligned}$$

Recall the geodesic equation

$$\frac{d^2 x^\mu}{d\sigma^2} + \Gamma_{\alpha\beta}^\mu \frac{dx^\alpha}{d\sigma} \frac{dx^\beta}{d\sigma} = 0.$$

Take $\mu = \theta$. Then we have

$$\ddot{\theta} + \Gamma_{\theta\theta}^\theta \dot{\theta}^2 + 2\Gamma_{\theta\phi}^\theta \dot{\theta} \dot{\phi} + \Gamma_{\phi\phi}^\theta \dot{\phi}^2 = 0.$$

The 2 follows by symmetry in the bottom two arguments of the Christoffel symbol. This must be the same as the corresponding Euler-Lagrange equation

$$\ddot{\theta} - \sin \theta \cos \theta \dot{\phi}^2 = 0,$$

where we have divided an above equation by 2. This yields

$$\Gamma_{\theta\theta}^\theta = 0, \quad \Gamma_{\theta\phi}^\theta = 0, \quad \Gamma_{\phi\phi}^\theta = -\sin \theta \cos \theta.$$

We can also rearrange the other equation as

$$\frac{2 \cos \theta}{\sin \theta} \dot{\theta} \dot{\phi} + \ddot{\phi} = 0.$$

This is the other part of the geodesic equation, giving

$$\Gamma_{\theta\theta}^{\phi} = 0, \quad \Gamma_{\phi\phi}^{\phi} = 0, \quad \Gamma_{\theta\phi}^{\phi} = \frac{\cos \theta}{\sin \theta} = \cot \theta.$$

The Christoffel symbols are thus

$$\Gamma_{\alpha\beta}^{\theta} = \begin{pmatrix} 0 & 0 \\ 0 & -\sin \theta \cos \theta \end{pmatrix}, \quad \Gamma_{\alpha\beta}^{\phi} = \begin{pmatrix} 0 & \cot \theta \\ \cot \theta & 0 \end{pmatrix}.$$

(b) Let $x^{\mu}(\phi) = (\pi/2, \phi)$, which is clearly the path along the equator. We compute the proper distance derivatives:

$$\frac{dx^{\theta}}{d\sigma} = 0, \quad \frac{dx^{\phi}}{d\sigma} = \frac{dx^{\phi}}{d\phi} \frac{d\phi}{d\sigma} = \dot{\phi}, \quad \frac{d^2 x^{\phi}}{d\sigma^2} = \frac{d}{d\sigma} \dot{\phi} = \ddot{\phi}.$$

Using the Christoffel symbols found in part (a), the geodesic equation reads

$$\frac{d^2 x^{\theta}}{d\sigma^2} - \sin \theta \cos \theta \left(\frac{dx^{\phi}}{d\sigma} \right)^2 = 0,$$

$$\frac{d^2 x^{\phi}}{d\sigma^2} + 2 \cot \theta \frac{dx^{\theta}}{d\sigma} \frac{dx^{\phi}}{d\sigma} = 0.$$

Plugging in our derivatives and $\cos \pi/2 = 0$, this reads

$$0 - 0 \sin \theta \left(\frac{dx^{\phi}}{d\sigma} \right)^2 = 0 \implies 0 = 0,$$

$$\ddot{\phi} + 2 \cot \theta \dot{\theta} \dot{\phi} = 0 \implies \ddot{\phi} = 0.$$

It suffices to show this final equality holds. Using the metric we have

$$1 = \left(\frac{d\theta}{d\sigma} \right)^2 + \sin^2 \theta \left(\frac{d\phi}{d\sigma} \right)^2.$$

If we take θ constant at $\pi/2$, then $\frac{d\theta}{d\sigma} = 0$ and $\sin \theta = 1$, so we obtain

$$\pm 1 = \frac{d\phi}{d\sigma}.$$

It follows that

$$\frac{d^2 \phi}{d\sigma^2} = \pm \frac{d}{d\sigma} 1 = 0.$$

Therefore, the statement follows.

Problem 2

(a) The path is $x^\mu = (\bar{\theta}, \phi)$. Parallel transport then gives

$$\frac{D}{d\sigma} V^\mu = \frac{dV^\mu}{d\sigma} + \Gamma_{\alpha\beta}^\mu \frac{dx^\alpha}{d\sigma} V^\beta = 0.$$

Note that

$$\frac{dx^\theta}{d\sigma} = 0, \quad \frac{dx^\phi}{d\sigma} = \dot{\sigma}.$$

Using the Christoffel symbols from 1.a, we have the equations

$$\begin{aligned} \frac{dV^\theta}{d\sigma} - \sin \bar{\theta} \cos \bar{\theta} \frac{dx^\phi}{d\sigma} V^\phi &= 0, \\ \frac{dV^\phi}{d\sigma} + \cot \bar{\theta} \frac{dx^\phi}{d\sigma} V^\theta &= 0. \end{aligned}$$

Here we have noted that since $x^\theta = \bar{\theta}$ is constant, $\frac{dx^\theta}{d\sigma} = 0$. Note that when $\bar{\theta}$ is fixed,

$$1 = \left(\frac{d\theta}{d\sigma} \right)^2 + \sin^2 \theta \left(\frac{d\phi}{d\sigma} \right)^2 = \sin^2 \bar{\theta} \left(\frac{d\phi}{d\sigma} \right)^2 \implies \frac{d\phi}{d\sigma} = \frac{1}{\sin \bar{\theta}}.$$

We then obtain $\frac{d\theta}{d\sigma} = 0$. Noting that $x^\phi = \phi$, applying this to the above equations yields

$$\begin{aligned} \frac{dV^\theta}{d\sigma} &= \cos \bar{\theta} V^\phi, \\ \frac{dV^\phi}{d\sigma} &= -\frac{\cos \bar{\theta}}{\sin^2 \bar{\theta}} V^\theta. \end{aligned}$$

To solve the system, differentiate the first equation to obtain

$$\frac{d^2 V^\theta}{d\sigma^2} = \cos \bar{\theta} \frac{dV^\phi}{d\sigma},$$

noting that $\cos \bar{\theta}$ is a constant. Plugging this into the second equation yields

$$\frac{1}{\cos \bar{\theta}} \frac{d^2 V^\theta}{d\sigma^2} = -\frac{\cos \bar{\theta}}{\sin^2 \bar{\theta}} V^\theta \implies \frac{d^2 V^\theta}{d\sigma^2} = -\cot^2 \bar{\theta} V^\theta.$$

This is solved by

$$V^\theta = \exp(\pm \sigma \cot \bar{\theta}) \xrightarrow{\mathbb{C} \rightarrow \mathbb{R}} \{\cos(\sigma \cot \bar{\theta}), \sin(\sigma \cot \bar{\theta})\}.$$

The most general form of V^θ is thus

$$V^\theta = C_1 \cos(\sigma \cot \bar{\theta}) + C_2 \sin(\sigma \cot \bar{\theta}).$$

We use this to solve for V^ϕ using the second equation. Computing the derivative yields

$$\frac{dV^\theta}{d\sigma} = -C_1 \cot \bar{\theta} \sin(\sigma \cot \bar{\theta}) + C_2 \cot \bar{\theta} \cos(\sigma \cot \bar{\theta}) = \cos \bar{\theta} V^\phi.$$

We obtain

$$V^\phi = \frac{1}{\sin \bar{\theta}} (C_2 \cos(\sigma \cot \bar{\theta}) - C_1 \sin(\sigma \cot \bar{\theta})).$$

As found previously,

$$\frac{d\phi}{d\sigma} = \frac{1}{\sin \bar{\theta}} \implies \sigma = \phi \sin \bar{\theta}.$$

We can substitute this in to obtain

$$\begin{aligned} V^\theta &= C_1 \cos(\phi \cos \bar{\theta}) + C_2 \sin(\phi \cos \bar{\theta}), \\ V^\phi &= \frac{1}{\sin \bar{\theta}} (C_2 \cos(\phi \cos \bar{\theta}) - C_1 \sin(\phi \cos \bar{\theta})). \end{aligned}$$

Write $\omega = \cos \bar{\theta}$ to simplify notation. Imposing the initial condition $V^\mu(0) = (a, b)$ yields

$$\begin{aligned} a &= C_1 \cos(0\omega) + C_2 \sin(0\omega) = C_1, \\ b &= \frac{1}{\sin \bar{\theta}} (C_2 \cos(0\omega) - C_1 \sin(0\omega)) = \frac{1}{\sin \bar{\theta}} C_2 \implies C_2 = b \sin \bar{\theta}. \end{aligned}$$

We thus obtain

$$\begin{aligned} V^\theta &= a \cos(\omega\phi) + b \sin \bar{\theta} \sin(\omega\phi), \\ V^\phi &= b \cos(\omega\phi) - \frac{a}{\sin \bar{\theta}} \sin(\omega\phi). \end{aligned}$$

(b) The initial condition gives $V^\mu(0) = (a, b)$. We compute $V^\mu(2\pi)$:

$$\begin{aligned} V^\theta(2\pi) &= a \cos(\omega 2\pi) + b \sin \bar{\theta} \sin(\omega 2\pi), \\ V^\phi(2\pi) &= b \cos(\omega 2\pi) - \frac{a}{\sin \bar{\theta}} \sin(\omega 2\pi). \end{aligned}$$

For us to simplify any further, we would need to know more about $\omega = \cos \bar{\theta}$. In general, this may differ from $V^\mu(0)$. In particular, if $\bar{\theta} = \pi/2$, then cosine goes to 0 and sin goes to 1. We then have

$$\begin{aligned} V^\theta(\phi) &= a \cos(0) + \frac{b}{1} \sin(0) = a, \\ V^\phi(\phi) &= b \cos(0) - \frac{a}{1} \sin(0) = b. \end{aligned}$$

Therefore, $V^\phi(2\pi) = (a, b) = V^\mu(0)$ when $\bar{\theta} = \pi/2$.

We do not need to compute this, since we saw in problem 1 that the path around the equator (which is precisely x^μ with $\bar{\theta} = \pi/2$) is a geodesic. Along the geodesic, the metric reduces to the same as flat space. Parallel transporting can then be viewed as parallel transporting along a straight line, which will not change the vector components.